

SafeRoadIndia

Analytics Dashboard

**Interactive Road Accident Analytics & Machine
Learning-Based Risk Prediction**

**Prepared by:
Shriyans Mohanty
(12400154)**

Training Organization: O7 Services, Jalandhar

Demo link: [SafeRoadIndiaAnalyticsDashboard](#)

1. Introduction

Injuries and deaths from road traffic remain among the leading causes of death and injury globally. To improve road safety and support sound decision-making, it is fundamental to understand accident patterns and the factors contributing to accidents.

SafeRoadIndia is an interactive analytics dashboard developed using Streamlit that enables users to explore Indian road accident data using visualizations and also predicts accident risk using a Machine Learning model.

The project combines data analytics, interactive visualization and machine learning into a single, user-friendly application.

2. Problem Statement

Big road accident data sets are full of insights, but difficult to analyze using traditional spreadsheets or static reports. Analysts and authorities need an interactive platform that:

- Look at accident trends
- Identify accident black spots
- Accident severity comparison under different conditions
- Picture accident patterns
- Predict accident risk from environmental and traffic conditions.

Without this system, it's time-consuming and less effective to derive meaningful insights from large datasets.

3. Proposed Solution

To tackle these challenges, SafeRoadIndia provides an interactive dashboard that can:

- Filter accident data by state, year and city
- Viewing accident trends with several graphs
- Displaying density of accident hotspots in India
- Cause and severity analysis
- Accident Risk Prediction with a Trained Machine Learning Model
- Explanation of the main factors contributing to the prediction.

The dashboard enables the users to quickly identify important patterns and facilitates data-driven analysis of road safety.

4. What is SafeRoadIndia?

SafeRoadIndia is an interactive analytics dashboard built on Streamlit for analyzing Indian road accidents.

It blends statistical analysis with machine learning to offer both descriptive and predictive insights.

Key Features

- Interactive Dashboard
- Dynamic Filters
- KPI Cards
- Accident Hotspot Map
- State & City Analysis
- Weather Analysis
- Road Type Analysis
- Accident Cause Analysis
- Year-wise Trend Analysis
- Day × Hour Heatmap
- Time-of-Day Distribution
- Machine Learning Risk Prediction
- Risk Gauge Visualization
- Explainable Risk Factors

5. Technology Stack



Folium

6. Dataset

Source: [Kaggle](#)

The dataset contains Indian road accident records from **2022 to 2025** and includes information such as

- State
- City
- Temperature
- Weather
- Visibility
- Road Type
- Traffic Density
- Number of Lanes
- Time
- Day of Week
- Accident Cause
- Accident Severity

7. Machine Learning Module

The dashboard includes a Machine Learning model that estimates accident risk based on user-selected parameters.

Input Parameters

- Temperature
- Hour
- Number of Lanes
- Weather
- Visibility
- Traffic Density
- Road Type
- Peak Hour
- Weekend
- Festival
- Traffic Signal

The predicted result is displayed through an intuitive **risk gauge** accompanied by the major contributing factors affecting the prediction.

8. Dashboard Visualizations

The dashboard provides several interactive visualizations including:

- Accident Hotspot Density Map
- Top Accident-Prone States
- Top Accident-Prone Cities
- Weather vs Accident Severity
- Road Type vs Accident Severity
- Accident Cause Analysis
- Year-wise Accident Trend
- Day × Hour Heatmap
- Time of Accident Distribution

These visualizations enable users to identify patterns and understand accident behavior across different conditions.

9. Results

The SafeRoadIndia dashboard successfully demonstrates how data analytics and machine learning can be integrated into a single platform for road accident analysis.

The project enables users to:

- Analyze accident trends interactively
- Explore accident hotspots
- Compare accident severity across different conditions
- Predict accident risk
- Understand the factors contributing to the predicted risk

The system provides a user-friendly interface suitable for educational, analytical, and demonstration purposes.

11. Conclusion

SafeRoadIndia is a working use case for data analytics and machine learning pertaining to road safety. The dashboard incorporates interactive visualizations and predictive analytics, converting complex accident datasets into meaningful insights that are easy to understand and explore.

The project improved practical skills in preprocessing data, visualization, dashboard development, and machine learning and laid a scalable foundation for future intelligent transportation and road safety applications.

12. Screenshots



